PROJECT PROPOSAL

Adaptive Focus Studio

Student Name	Samuel Raju Parisa
Course	Flutter Build Studio – Project One
Academic Pathway	Graduate
Project Type	Individual work requested; approval pending
Date	October 18, 2026

1. Introduction

My project is called Adaptive Focus Studio. It is a Flutter mobile application that helps students plan and keep track of their study sessions. Users can create focus presets, start a timer, save completed sessions, and check their previous activity. The app will store its data on the device so it can be used without an internet connection.

2. Problem Statement

Students sometimes plan to study for a certain amount of time but do not keep track of how long they actually focus. Without a record of previous sessions, it can be difficult to choose a useful duration for the next session. This project will put the timer, saved sessions, and a simple progress summary in one application.

3. Objectives

Allow users to create, view, edit, and delete focus presets.

Allow users to start, pause, resume, finish, or reset a focus session.

Save focus presets and session records in a local SQLite database.

Show session history with a filter and a summary such as average focus time.

Recommend a focus duration using the user's previous completed sessions.

Keep the main features working offline and preserve data after the app is closed.

4. Scope

The application will focus on individual study sessions. It will not require user accounts, a cloud database, or a remote AI service. The main features will be preset management, a focus timer, session history, settings, and a local duration recommendation.

5. Main Screens

Dashboard / Preset Catalog: Shows saved focus presets and a short activity summary. The user can start a session, edit a preset, or add a new one.

Preset Editor: Lets the user enter a title, category, target duration, and optional notes. The form will check that the input is valid.

Timer / Session Detail: Shows the selected preset and timer. The user can start, pause, resume, finish, or reset the session.

Session History: Shows previous sessions and lets the user filter records. It will display a summary such as average completed focus time.

Settings: Stores preferences such as theme, default focus duration, and notification preference.

6. User Stories

1. As a student, I want to create and edit focus presets so I can reuse my usual study durations.

Acceptance criteria: a preset can be saved and edited; blank titles and invalid durations show an error; deleting a preset asks for confirmation.

2. As a student, I want to control a focus timer so I can record a study session.

Acceptance criteria: the timer has start, pause/resume, finish, and reset/cancel actions; a finished session is saved once; invalid actions are handled clearly.

3. As a student, I want to review previous sessions so I can understand my study habits.

Acceptance criteria: history can be filtered; a summary is calculated from saved records; an empty-history message is shown when there are no sessions.

7. Database Design

I plan to use two related SQLite tables:

Table	Fields	Purpose
FocusPreset	id, title, category, targetMinutes, notes, createdAt, updatedAt	Stores reusable focus plans.
FocusSession	id, presetId, startedAt, endedAt, actualSeconds, status, reflection	Stores each session and links it to a preset.

Both tables will support create, read, update, and delete operations. FocusSession will have a foreign key to FocusPreset. I plan to enable foreign-key enforcement. Deleting a preset will show a warning that linked sessions will also be deleted, followed by a cascade-delete policy. I will test this behavior.

SQLite will store the main records. SharedPreferences will store small settings such as theme and default duration. The app should work without an internet connection.

8. Local Smart Helper

The smart helper will recommend a focus duration using the user's previous completed sessions. It will show a short explanation for the recommendation, and the user can apply or ignore it. If there is no session history, the app will show a default suggestion instead of pretending it has personalized information.

For the Graduate pathway, I will compare the helper with a fixed 25-minute baseline using at least five held-out cases. I will calculate mean absolute error in minutes and report the results, including cases where the helper does not perform better. AI_FEATURE.md will explain the rule, baseline, test cases, results, and limitations. The feature will run locally and will not call a cloud AI API.

9. Technology and Implementation

Flutter and Dart for the mobile app.

SQLite with sqflite for focus presets and session records.

SharedPreferences for small user settings.

Riverpod for state management.

Separate UI, domain, and data code, with repositories and dependency injection.

Local notifications, a useful file export/import feature, and purposeful animations.

At least 10 unit tests, widget tests, integration tests, and a coverage report.

10. Testing Plan

Test creating, viewing, editing, and deleting presets and sessions.

Test invalid input, including blank titles and out-of-range durations.

Test timer start, pause/resume, finish, reset, and repeated finish attempts.

Test the foreign-key relationship and parent-delete policy.

Restart the app to confirm records and settings remain saved.

Test history filters and summary calculations using known sample records.

Test helper cases: normal history, no history, invalid values, boundaries, an outlier, and a counterexample.

Test on two Android devices or emulators and use peer feedback to make and test one design change.

11. Initial Wireframe Sketches

These are my first layout plans. I will refine them before implementation.

Dashboard

[Title / Settings]
[Activity summary]
[Preset list]
[Start] [Edit] [Add preset]

Preset Editor

[Title field]
[Category]
[Duration slider]
[Notes]
[Save] [Cancel]

Timer

[Preset name]
[Timer display]
[Start / Pause / Resume]
[Finish] [Reset]

History

[Filter]
[Average focus time]
[Session list]
[Session details]

Settings

[Theme]
[Default duration]
[Notifications]
[Export data]

12. Work Plan

Date	Planned work
October 18	Submit proposal, wireframes, repository and board links, and signed statement. Wait for instructor approval.
October 19–21	After approval, build the main screens and navigation and start the database.
October 22–24	Complete CRUD, timer workflow, settings, history, and local helper.
October 25–26	Complete tests, device checks, helper evaluation, and a peer-based design revision.
October 27–28	Finish report, presentation, evidence, APK, and final submission.

13. Project Links and Approval

I am requesting permission to complete this project individually. The project instructions describe group development, so I will wait for written instructor approval before starting implementation. I will use the required project board and keep real, dated progress evidence as directed by the instructor.

Student signature: Samuel Raju Parisa

Date: 10/18/26

Instructor approval / comments: _____

14. References

Flutter documentation: https://docs.flutter.dev/

SQLite documentation: https://www.sqlite.org/docs.html

Flutter Build Studio Project One brief and topic description provided for this course.